HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering

Batch: 22nd
Date: 07/12/2025

Mid-Term Examination

Semester: Fall 2025
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions: $5 \times 6 = 30$

1. (a) Explain the general principles of software engineering. 3
   (b) List the disadvantages of legacy software. 3

2. (a) Draw all four process-flow-diagrams in generic process model. 3
   (b) Describe stage pattern, task pattern and phase pattern. 3

3. (a) Explain component-based development process model. 3
   (b) Create a task set for a large and complex software project. 3

4. (a) Explain the human factors involved in agile process. 3
   (b) What is Agility? List all Agility principles. 3

5. (a) Draw a table for Extreme programming practices. 3
   (b) Describe Industrial Extreme Programming (IXP). 3

6. (a) Define product backlog and burn down chart in Scrum. 3
   (b) Define the role of a Scrum Master. 3

7. (a) Create a table comparing different agile techniques. 3
   (b) Briefly explain the stages involved in DevOps approach. 3

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CSE423: Simulation and Modeling

Batch: 22nd | Semester: Fall 2025

Time: 1 Hour 30 Minutes

Date: 11-12-2025 | **Mid-term Examination** | Total Marks: 30

Answer any **five** questions of the followings. $5 \times 6 = 30$

1. (a) Define any three terminologies of system simulation with an example. 3
   (b) Draw the flowchart of steps involved in simulation study. 3

2. (a) What are the advantages of simulation over analytical modeling? 3
   (b) What are the common problems faced during verification and validation? 3

3. (a) Differentiate between continuous and discrete systems. 2
   (b) State the steps of a discrete event simulation model. 4

4. (a) How is probability involved in discrete event simulation? 2
   (b) Illustrate the four activities that occur inside the single-server queueing system. 4

5. (a) Briefly explain different types of perspectives for modeling of a system. 3
   (b) Construct a diagram for arrival routine in queueing system. 3

6. (a) In an M|M|1 queue if $\lambda = 9$, M= 12 find the W and W_q. 2
   (b) In time series validation: 4
   Real: [3, 4, 8, 10, 13]
   Simulation: [2, 3, 7, 9, 11]
   Calculate mean Absolute Percentage Error (MAPE) and check the validity.

7. In an M/M/1/K queuing system with a single server, arrival rate $\lambda = 3$/hr, service rate $\mu = 4$/hr, system capacity k=4. Find the steady possibilities P(n), Blocking possibility P_k, Average number in system L and effective arrival rate λ_{eff} and loss Rate. 6

HAMDARD UNIVERSITY BANGLADESH

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Department of Computer Science and Engineering

CSE 469 : Digital Image Processing

Batch: 22nd | Semester: Fall 2025

Date: 15/12/2025 | **Midterm Examination** | Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

1. (a) Define Digital Image. Mention some application areas of Digital Image Processing. **2**

   (b) Briefly describe the fundamental steps in Digital Image Processing. **4**

2. (a) How much memory is needed to store a digital image? Explain with an example. **2**

   (b) Explain a Simple Image Formation Model. **4**

3. (a) Write short notes on - **3**

   (i) Neighbors of a Pixel and

   (ii) Adjacency of a Pixel.

   (b) What is the ambiguity problem associated with 8-adjacency and how can it be resolved? Explain using an example. **3**

4. (a) What is Histogram of an Image? Mention the obejectives or goal of Histogram Equalization. **2**

   (b) Suppose that a 2-bit image ($L = 4$) of size 32×32 pixels ($MN = 1024$) has the following intensity distribution, where the intensity levels are integers in the range $[0, L-1] = [0, 3]$. **4**

Intensity r_k	*Pixel Count* n_k
$r_0 = 0$	400
$r_1 = 1$	320
$r_2 = 2$	200
$r_3 = 3$	104

   Now, compute the equalized histogram values using the histogram equalization transformation function and illustrate the original histogram and equalized histogram.

5. (a) Use the following 3×3 mask to perform the convolution process on the shaded pixels in the 5×5 image below. Write the filtered image pixel values. **3**

0	1/4	0
1/4	1/2	1/4
0	1/4	0

3×3 mask

30	40	50	70	90
40	50	80	60	100
35	255	70	0	120
30	45	80	100	130
40	50	90	125	140

5×5 image

(b) Given the following 1-D image strip: 3

5	5	4	3	2	1	0	0	5	0	0	0	1	3	1	0	0	0	7	7	7

Compute the first and second derivative and using your results, briefly explain how the first and second derivatives respond to thick edges, thin edges and gray-level steps.

6. (a) Explain why do we need Fourier transform in digital image processing. 2
   (b) Distinguish between Spatial and Frequency domain filtering techniques. 2
   (c) Mention or illustrate the steps involved in filtering in the frequency domain. 2

7. (a) Explain different types of Smooting (Lowpass) frequency domain filters. 3
   (b) Differentiate between Smoothing and Sharpening filtering techniques. 3